

problem4

May 3, 2026

0.1 2.4 Cart-pole swing-up

(a) Please see `cartpole_swingup.py`.

(b)

Using a second order Taylor expansion of the stage cost $c(\tilde{s}_k, \tilde{u}_k)$ around $(\bar{s}_k, \bar{u}_k)$:

$$c(\tilde{s}_k, \tilde{u}_k) = c(\bar{s}_k, \bar{u}_k) + \underbrace{\left(\frac{\partial c}{\partial s} \Big|_{(\bar{s}_k, \bar{u}_k)} \right)}_{=q_k}^\top \tilde{s}_k + \underbrace{\left(\frac{\partial c}{\partial u} \Big|_{(\bar{s}_k, \bar{u}_k)} \right)}_{=r_k}^\top \tilde{u}_k + \underbrace{\frac{1}{2} \tilde{s}_k^\top \frac{\partial^2 c}{\partial s^2} \Big|_{(\bar{s}_k, \bar{u}_k)}}_{=Q} \tilde{s}_k + \underbrace{\frac{1}{2} \tilde{u}_k^\top \frac{\partial^2 c}{\partial u^2} \Big|_{(\bar{s}_k, \bar{u}_k)}}_{=R} \tilde{u}_k$$

hence $\boxed{q_k = Q(\bar{s}_k - s_{\text{goal}})}$ and $\boxed{r_k = R\bar{u}_k}$.

Same for the terminal cost $c_N(\tilde{s}_N)$:

$$c_N(\tilde{s}_N) = c_N(\bar{s}_N) + \underbrace{\left(\frac{\partial c_N}{\partial s} \Big|_{\bar{s}_N} \right)}_{=q_N}^\top \tilde{s}_N + \underbrace{\frac{1}{2} \tilde{s}_N^\top \frac{\partial^2 c_N}{\partial s^2} \Big|_{\bar{s}_N}}_{=Q_N} \tilde{s}_N$$

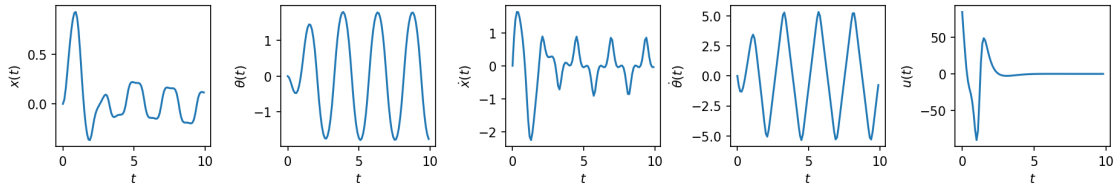
hence $\boxed{q_N = Q_N(\bar{s}_N - s_{\text{goal}})}$.

(c) Please see `cartpole_swingup.py`.

(d) Please see `cartpole_swingup.py`.

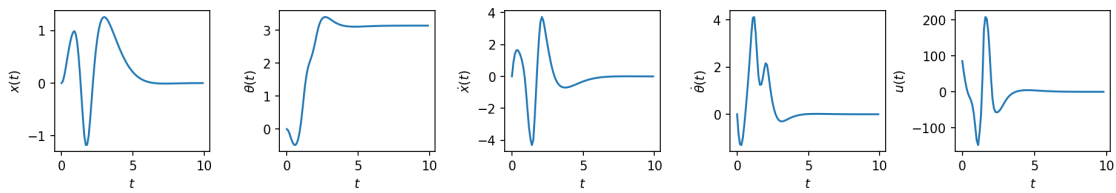
```
[1]: from IPython.display import Image, display
      print("Open-loop:")
      display(Image(filename='cartpole_swingup_ol.png'))
```

Open-loop:



```
[2]: print("Closed-loop:")
      display(Image(filename='cartpole_swingup_cl.png'))
```

Closed-loop:



While the open-loop is consistently oscillating, the closed-loop is progressively damped towards convergence as expected.